Dear Applicant,

Congratulations—you have made it to the Riverty Interview Challenge!

Why do we ask you to complete this challenge?

This challenge helps us get a more holistic picture of your skills, your way of thinking, and how you approach real-world problems. It complements the interview by giving you the opportunity to demonstrate your strengths in a practical and creative way.

We also want to be transparent about what we value and how we assess solutions. By sharing this challenge with you, we aim to create a fair and engaging experience that allows you to showcase your ideas, your technical approach, and your personal style. Ultimately, this challenge is designed to give you the space to shine and to start a meaningful conversation during the interview.

What do we expect from you as preparation?

You should be able to present your solution in 15–20 minutes during the interview. The format of your presentation is up to you. Of course, it can help to create a small slide deck to support your presentation, but remember we want to get to know you, not only your slides.

Problem Description

Over the last couple of years, AI has become a major driver of innovation. More and more corporations see an advantage in leveraging it to become more productive. While a technical definition of productivity is often rather complicated, it is economically easy to describe: create more with the same number of engineers.

This is also true for Riverty. However, we like to take on the challenges set by management and find creative and innovative technical solutions to succeed—while learning and improving ourselves at the same time.

One of these challenges is making our legal team more efficient. While there is a lot of talk about agents like Claude being able to help, in day-to-day operations they often fail due to the sheer size and complexity of the environments they have to operate in.

Technical Outline

Our legal colleagues currently leverage a variety of tools, but the main workload still consists of organizing and reviewing contracts, comparing them, and ensuring they comply with our corporate guidelines and represent the company's best interests.

Most documents reside on SharePoint and are in PDF format. Some of them are scans of really old, handwritten contracts that exist only as JPEG files. Additionally, there is a contract storage solution with a RESTful API. You may solely use it to store an additional copy of a contract in a legally compliant manner.

Riverty follows two guiding principles when it comes to technology choices. First, we evaluate best-of-breed solutions and what they offer. Second, as we are a Microsoft shop, we compare these solutions against what Microsoft provides.

As a result, we currently often use the following technologies:

- Azure
- Azure Foundry / Google Gemini
- Containerized environments
- Python with FastAPI and multiple libraries
- Terraform for infrastructure provisioning
- Frontend written in React and TypeScript
- LangChain and LangGraph for agents

The Task

Our legal colleagues regularly need to compare contracts and identify, for example, contracts that do not contain a certain text passage or contracts in which a company name needs to be updated.

So far, this is semi-manual work: they have to open the documents, search for the relevant content, and verify the results themselves.

They approached us to ask whether AI could help with this task, as it is purely repetitive work that nobody enjoys doing.

Your task is to come up with the following:

Presentation outlining:

- A solution explaining how you would tackle the challenge
- An overall technical solution you would present to our team and how it could be implemented
- A basic explanation of the infrastructure you would leverage
- A concept for helping our colleagues adopt the solution and ensure they get the most value out of it
- Anything else you feel is relevant to understanding your approach to solving the challenge

Basic implementation demonstrating the foundation of the concept, including:

- A basic pipeline showing how you would analyze the documents and make them accessible
- A basic front-end demonstrating how the solution can be used
- Anything you feel is necessary to demonstrate the end-to-end workflow

As a team, we encourage any form of AI usage, as long as we understand what is happening. As our future teammate, we encourage you to do the same.